■ Rationalize

Rationalize[x] takes Real numbers in x that are close to rationals, and converts them to exact Rational numbers.

Rationalize[x, dx] performs the conversion whenever the error made is smaller in magnitude than dx.

Example: Rationalize[3.78] $\longrightarrow \frac{189}{50}$. ■ Rationalize[N[Pi]] $\longrightarrow$ 3.14159 does not give a rational number, since there is none "sufficiently close" to N[Pi]. ■ A rational number p/q is considered "sufficiently close" to a Real x if $|p/q - x| < c/q^2$, where c is chosen to be 10^{-4}. See page 328. ■ See also: Chop, Round.

Web sample page from The *Mathematica* Book, Second Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-51502-4; softcover ISBN 0-201-51507-5). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; http://www.wolfram.com/; 1-800-441-6284.

© 1991 Wolfram Research, Inc. Permission is hereby granted for web users to make one paper copy of this page for their personal use. Further reproduction, or any copying of machine-readable files (including this one) to any server computer, is strictly prohibited.